

Acme Engineering Handbook

Platform Engineering Standards — Edition 2023

1. Architecture Overview

The Acme platform is a microservices architecture deployed on EKS (managed Kubernetes on AWS). Services communicate via gRPC for internal calls and REST for external APIs. The primary datastore is PostgreSQL 14, with Redis for caching and Kafka for event streaming. Object storage uses S3 with cross-region replication.

All services must expose health endpoints at `/healthz` and `/readyz`, and a Prometheus metrics endpoint at `/metrics`. Service discovery uses Kubernetes native DNS. Configuration is managed via Helm values and ConfigMaps.

2. Deployment Process

All deployments go through GitHub Actions. Production deploys require two approvals and must occur between 10:00 and 18:00 IST on weekdays. The deployment freeze window is the last Friday of each month and the final two weeks of December. Rollback must be possible within 5 minutes.

Deployments use a blue-green strategy. Canary deploys are optional and available for high-risk services. Progressive rollouts are managed through Argo Rollouts.

3. On-Call Rotation

Each team maintains a primary and secondary on-call rotation with weekly shifts. PagerDuty is the source of truth for escalation. Sev-1 incidents require a response within 15 minutes; Sev-2 within 1 hour. Postmortems are mandatory for all Sev-1 and Sev-2 incidents and must be filed within 5 business days.

On-call compensation increases to \$350 per week. Comp time is provided at 1 hour per 4 hours of after-hours work. Managers must ensure on-call engineers have a light project load during rotation.

4. Code Review Standards

Every pull request requires at least one approving review from a code owner. PRs touching authentication, payments, or data deletion require two approvals. Maximum PR size is 400 lines of diff. All PRs must pass linting, unit tests, and integration tests before merge.

A new architectural review process applies to PRs that introduce a new service, datastore, or third-party dependency. Reviews are conducted by the Architecture Guild on a weekly cadence.

5. Observability

All services emit structured JSON logs to the central ELK stack. Metrics are scraped by Prometheus every 15 seconds and visualized in Grafana. Distributed traces are captured via OpenTelemetry with a 5 percent sampling rate in production and 100 percent in staging.

Service Level Objectives (SLOs) are defined for all tier-1 services. Error budgets are tracked monthly. Alerts are tied to SLO burn rates rather than raw thresholds.

6. Testing Standards

Unit test coverage must be at least 80 percent per service. Integration tests must cover all public endpoints. Contract tests are required for service-to-service dependencies. End-to-end tests run on every merge to main and nightly on staging.

7. Security

All production secrets are stored in AWS Secrets Manager. Static analysis (SAST) and dependency scanning run on every PR. Container images are scanned with Trivy before promotion. Penetration tests are conducted annually by an external firm.